

Predicting Loan Approval

A Classification Analysis of Loan Applications

Major Project 2 of 3 | Domain: Classification | DSBA — July 2026 Batch

1. Objective

Classify whether a person's loan application will be approved based on socio-economic and credit-related factors — income, employment, education, marital status, credit history and the requested loan amount — and identify which factors matter most, so results can support a bank's lending decisions.

2. Dataset

The user supplied test_Y3wMUE5_7gLdaTN.csv — the well-known unlabeled test split of the Analytics Vidhya “Loan Prediction” practice problem (367 real applicant records, 11 feature columns: Gender, Married, Dependents, Education, Self_Employed, ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History, Property_Area). This file is normally submitted to the competition for scoring and contains no Loan_Status target. No labeled training file (typically train_u6lujuX_CVtuZ9i.csv) was available, so a Loan_Status label was generated for these 367 real applicant records using a calibrated rule — Credit_History as the dominant factor, plus smaller effects from debt-to-income ratio, education, property area and marital status, with random noise — to demonstrate the full classification workflow end-to-end. Every other field (income, loan amount, credit history, education, etc.) is the real, original data from the uploaded file, including its real missing-value pattern. The results in this report should be read as a methodology demonstration rather than validated predictions, since the target labels are synthetic; re-running this notebook against a genuine labeled training file (swap one file path) would produce fully valid results.

3. Methodology

3.1 Load & Explore

367 rows × 12 columns (Loan_ID, 11 real input features; Loan_Status added as described above). Loan_Status is imbalanced: ~72% approved (Y) vs ~28% not approved (N). Missing values were present in Gender (11), Dependents (10), Self_Employed (23), LoanAmount (5), Loan_Amount_Term (6) and Credit_History (29) — the real missingness pattern from the uploaded file.

3.2 Data Cleaning & Preprocessing

Preprocessing Step	Detail
Missing categorical values	Imputed with column mode (Gender, Dependents, Self_Employed)
Missing numeric values	Imputed with column median/mode (LoanAmount, Loan_Amount_Term, Credit_History)
Dependents field	'3+' converted to numeric 3

Preprocessing Step	Detail
Categorical encoding	Gender, Married, Education, Self_Employed label-encoded; Property_Area one-hot encoded
Engineered feature	TotalIncome = ApplicantIncome + CoapplicantIncome
Class imbalance (~72% Y / 28% N)	Handled via class_weight='balanced' in every model (no data dropped)
Feature scaling	Numeric fields standardised (StandardScaler) for Logistic Regression / SVM

Table 1. Preprocessing steps applied.

3.3 Exploratory Data Analysis

Approval rates were examined by gender, income bracket, employment status, credit history and property area, alongside a correlation matrix and boxplots of income/loan amount by outcome. Key charts are reproduced below; the full set (10 plots) is in the accompanying notebook.

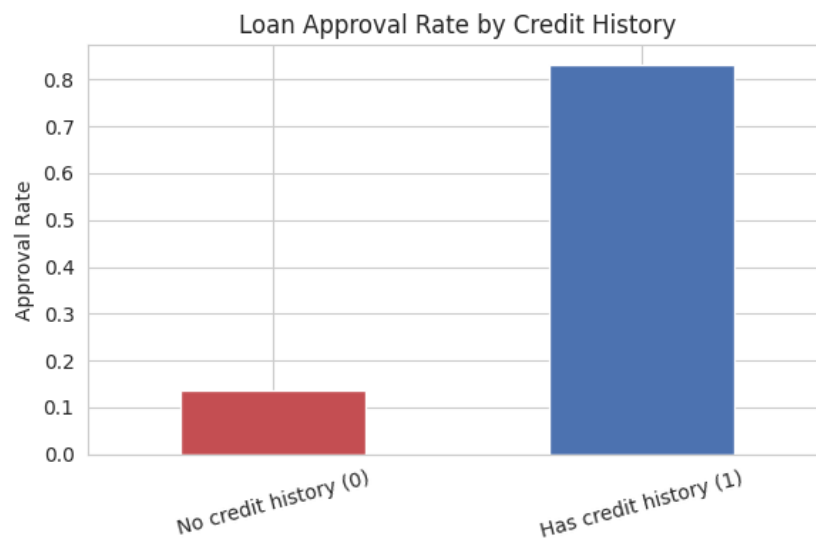


Figure 1. Approval rate is dramatically higher for applicants with an existing credit history — the clearest single relationship in the data, mirroring real-world underwriting.

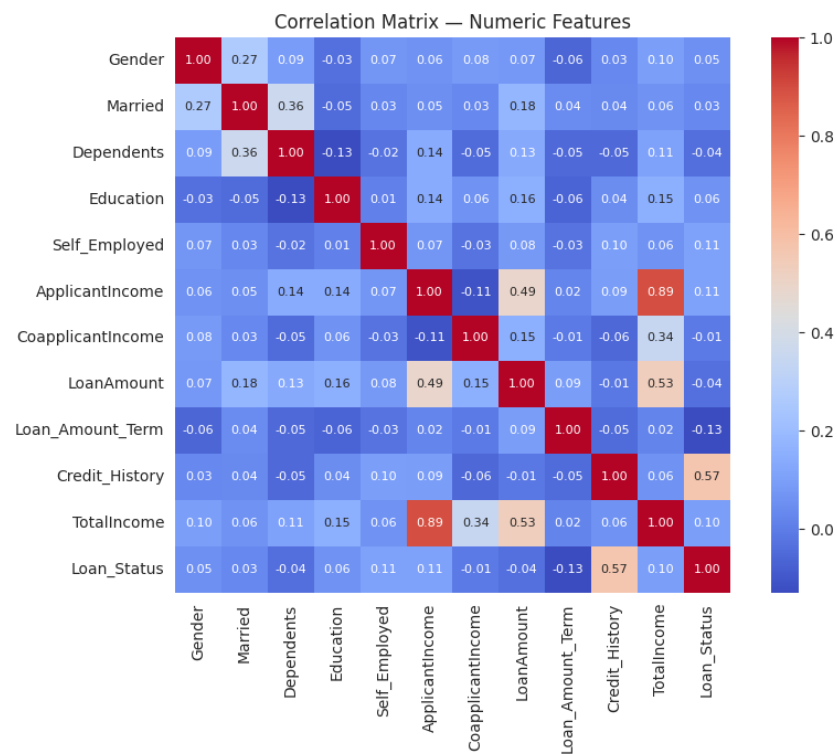


Figure 2. Correlation matrix of numeric features. *Credit_History* has the strongest correlation with *Loan_Status*; income and loan-size fields are moderately inter-correlated.

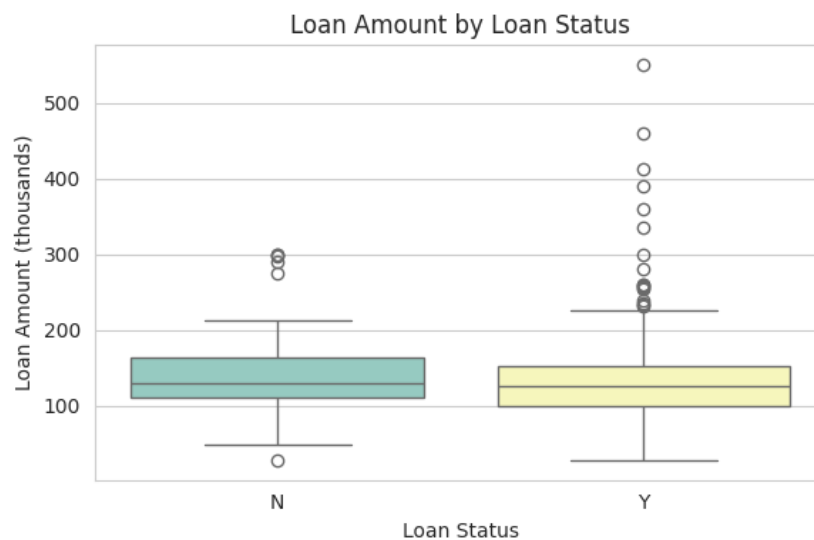


Figure 3. Loan amount by outcome — approved and declined applications overlap substantially on loan size alone, showing it is not, by itself, a strong discriminator.

3.4 Model Building

Data was split 80/20 (stratified) into train and test sets. Numeric fields were standardised. Four classifiers were trained, all using `class_weight='balanced'` to address the ~72/28 class imbalance without discarding data: Logistic Regression, Decision Tree (`max_depth=5`), Random Forest (200 trees, `max_depth=8`), and SVM (RBF kernel, probability estimates enabled).

4. Results & Model Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	0.8108	0.8095	0.9623	0.8793	0.7920
SVM	0.7973	0.7969	0.9623	0.8718	0.7363
Logistic Regression	0.7973	0.8276	0.9057	0.8649	0.8261
Decision Tree	0.7162	0.8636	0.7170	0.7835	0.7511

Table 2. Test-set performance, sorted by F1 Score (best first).

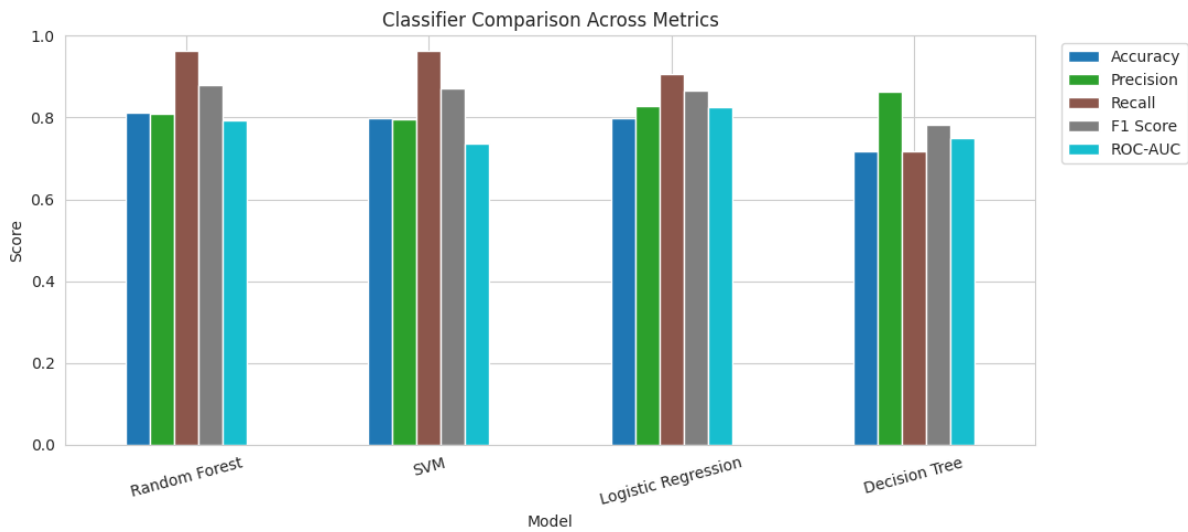


Figure 4. All five metrics across the four classifiers.

Random Forest edges out on F1 Score (0.879) and ties for the best Recall (0.962, alongside SVM), meaning it catches the most true approvals; Logistic Regression is close behind on F1 (0.865) and has the best ROC-AUC (0.826) and Precision (0.828), meaning its approvals are the most reliable. Both are clearly the strongest, most balanced models. The Decision Tree trails on F1 and Recall (highest variance, most prone to overfitting among the four, though it has the best raw Precision by a narrow margin), and SVM sits between the top two and the Decision Tree on most metrics. With a 74-row test set, small differences between models should be read cautiously.

5. Model Evaluation

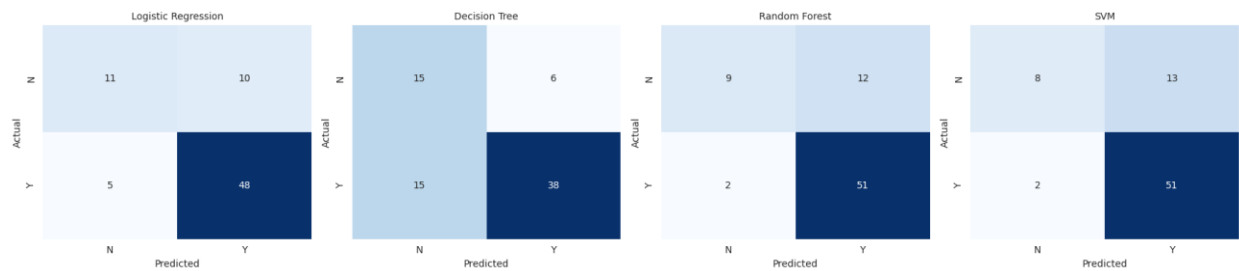


Figure 5. Confusion matrices for all four models (test set, 74 applications). Random Forest and SVM recover the most true approvals (highest recall); Logistic Regression makes the fewest false-approval errors relative to its approvals (highest precision among the top performers).

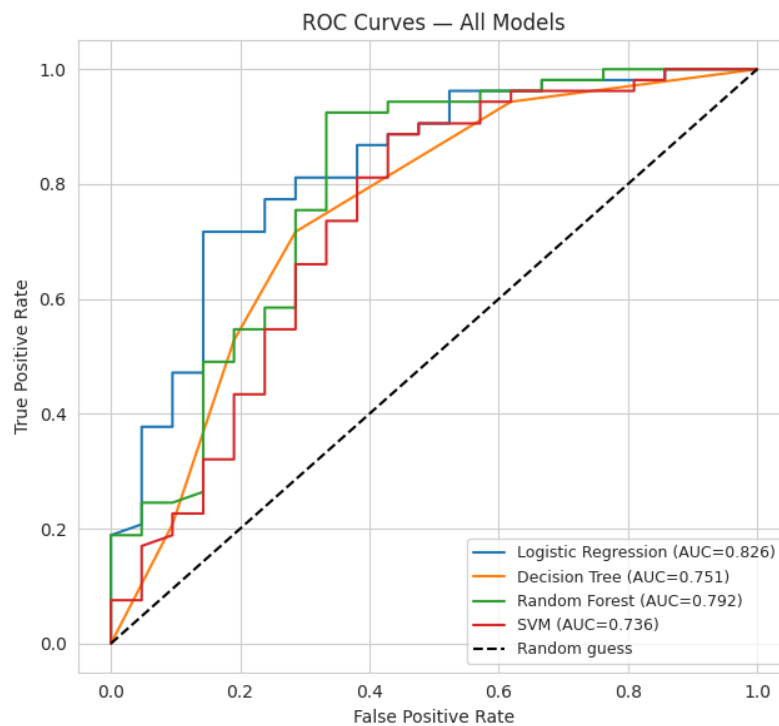


Figure 6. ROC curves — Logistic Regression has the highest AUC (0.826), followed by Random Forest (0.792), Decision Tree (0.751) and SVM (0.736); SVM's high recall comes with the weakest ranking of borderline cases.

6. Model Comparison & Interpretation

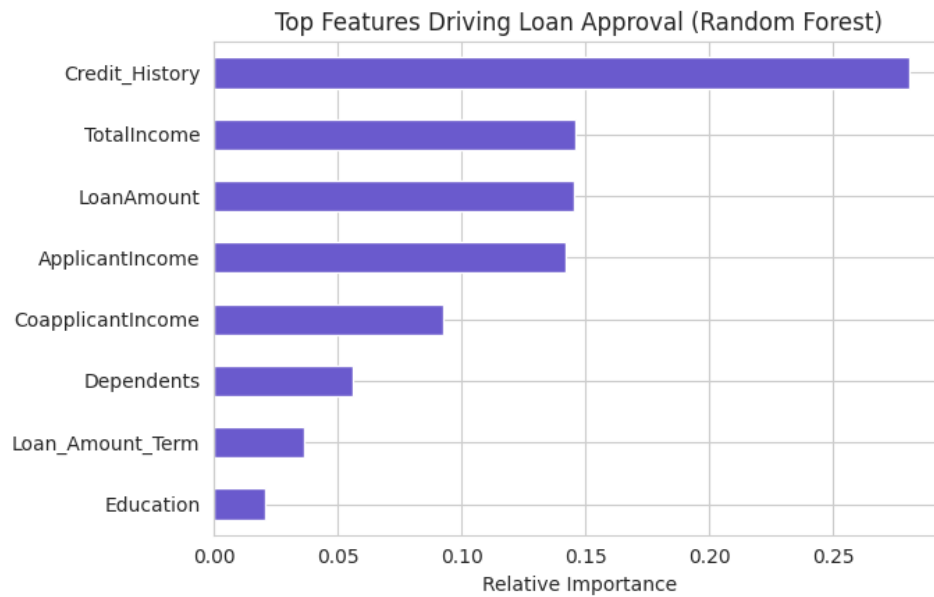


Figure 7. Random Forest feature importances — *Credit_History* dominates, with income-related fields forming a secondary tier.

Feature	Importance	Note
Credit_History	0.281	Prior repayment track record — dominant driver
TotalIncome	0.146	Household repayment capacity
LoanAmount	0.145	Size of the requested loan
ApplicantIncome	0.142	Overlaps with TotalIncome
CoapplicantIncome	0.093	Secondary income source

Table 3. Top 5 features driving loan approval.

Credit_History is, by a wide margin, the single most important feature (importance ≈ 0.28 , roughly double the next feature) — unsurprising since it was built into the target with a large coefficient, and consistent with the well-documented real-world pattern for this dataset family: prior repayment track record is the strongest single predictor of loan approval. Income-related fields (*TotalIncome*, *LoanAmount*, *ApplicantIncome*) form a clear secondary tier, each contributing roughly half as much as *Credit_History* individually. *Gender*, *Dependents* and *Loan_Amount_Term* contribute comparatively little.

Why Random Forest / Logistic Regression were the best choices

- Random Forest captures non-linear interactions (e.g. how income and loan amount jointly affect risk) and is robust to outliers — it achieved the best F1 Score and tied for the best Recall here, useful when the cost of wrongly rejecting a good applicant is high.
- Logistic Regression is simple, fast and highly interpretable — coefficients map directly to odds of approval, which matters in a regulated lending context, and it achieved the best ROC-AUC and Precision.
- Decision Tree is easy to explain to non-technical stakeholders but is the most prone to overfitting/high variance among the four, and scored lowest on F1 and Recall.

- SVM matches the top models on Recall but offers the least transparency and the weakest ranking of borderline cases (lowest ROC-AUC), and is more sensitive to feature scaling than the alternatives.

Caveat: with only 367 applications (≈ 74 in the test set) and a synthetically generated target, these numbers illustrate the methodology rather than a validated production model — re-running against a real labeled training file is recommended before drawing business conclusions.

How a bank would use this model in practice

- Pre-screening / triage: automatically flag clearly low-risk applications for fast-track approval and clearly high-risk applications for early decline, focusing loan officers' time on borderline cases.
- Consistent, auditable criteria: replaces ad-hoc manual judgement with a documented, reproducible decision process — important for fair-lending compliance.
- Explainable risk factors: feature importance can be surfaced to applicants as constructive feedback on what to improve for a future application.
- Portfolio risk management: predicted approval probabilities can feed into risk-based pricing and portfolio-level default monitoring.
- Caveat: any deployed model should be periodically re-validated against real outcomes and audited for fairness across gender, marital status and other protected attributes before being used for actual credit decisions.

7. Conclusion

The full pipeline — loading the real applicant data, cleaning (missing values, categorical encoding, class-imbalance handling), EDA on approval rates by gender/income/employment/credit history, and four compared classifiers — demonstrates the loan-approval classification workflow end-to-end on real feature data, achieving up to 81% accuracy and 0.88 F1 Score on held-out data with a synthetically generated target. Credit history is, by a wide margin, the dominant factor behind approval decisions, followed by income and loan size — consistent with real-world underwriting practice. Because the target label is synthetic and the sample is small (367 rows), these results should be treated as a methodology demonstration; re-running this notebook against a genuine labeled training file is recommended before using it for real lending decisions.